SYSTEM ARCHITECTURE & TECHNICAL SPECIFICATION WHITEPAPER

PashuRakshak (पशुरक्षक) - Animal Health & Outbreak Surveillance Platform

An End-to-End Clinical Screening, Geospatial Outbreak Sensing & Coordinated Epidemic Defense System

Problem Statement ID PS 26128 (Govt of Maharashtra)

Platform Core Go 1.22+ / Triple SQLite WAL

Clinical Decision Engine viggovet/viggoVet-Reasoning-20B

Document Edition v3.4.0 Production Release

1. Problem Statement Alignment (PS ID 26128)

🏛️ Official Problem Statement 26128 • Government of Maharashtra & MSIS

Title: Efficient systems for early detection, prevention, and management of livestock diseases and animal health issues **Description:** Livestock owners, field veterinarians, para-veterinary workers and government departments often lack a unified, realtime mechanism to identify emerging animal-health risks at the village, block and district levels. Disease symptoms may be reported late, diagnostic facilities may be distant, vaccination and treatment histories may be incomplete, and information from farms, veterinary dispensaries, laboratories, vaccination drives and surveillance programmes may remain fragmented. These gaps can delay containment, increase livestock mortality and productivity loss, raise the risk of zoonotic transmission, and affect farmers' incomes. The challenge is to create a practical system that enables early warning, rapid reporting, risk assessment, preventive action, referral and coordinated response.

The **PashuRakshak** platform is engineered as a direct solution to this mandate, uniting dairy farmers, pastoralists, licensed veterinary doctors, dispensaries, NGOs, and state authorities into an interconnected, real-time epidemiological grid.

2. Multi-Role Ecosystem & Geospatial Entity Mapping

To address fragmentation between field workers and clinical facilities, the platform partitions functionality across four primary user classes:

🚜 Livestock Farmers & Dairy Owners (`farmer`)

Registers farm name, village (गावाचे नाव), taluka, district, livestock types (Cattle, Buffalo, Goat, Sheep, Poultry, Swine), and herd size. Features livestock video/photo screening, real-time outbreak SOS radar, recovery reporting (`POST /api/outbreaks/report-recovery`), and nearby clinic directory.

🐾 Companion Pet Parents (`pet_owner`)

Registers address, city, district, and pincode. Features instant lesion & motion gait screening, broad-spectrum contagion alerts (e.g. Canine Parvovirus, Feline Panleukopenia), and one-tap doctor second-opinion requests.

🩺 Licensed Veterinarians & Clinics (`vet`)

Dedicated Doctor Decision Hub: Case Review Queue to inspect doubtful screenings, verify diagnoses, issue ℞ prescriptions, manage live clinic operating hours, broadcast current field visiting locations, and post leave notices.

📦 NGOs & Government Dispensaries (`ngo` / `gov`)

Real-time emergency inventory management for critical vaccines (FMD, Lumpy Skin, PPR), broad-spectrum antibiotics, antiseptics, and PPE kits with automated low-stock warnings and official circular broadcasting.

3. Dynamic Outbreak Sensing, Vet Consensus & Resolution Engine

Unlike static alert placeholders, PashuRakshak incorporates a dynamic epidemiological state engine:

🚨 Rule 1: Veterinarian Consensus Trigger

When ≥ 3 licensed veterinarians confirm the same infectious disease (e.g. FMD, Lumpy Skin, PPR, Parvo, Anthrax) within a

district, the system automatically flags a **Verified Outbreak** (`status = 'active'`, `severity = 'CRITICAL'`) and dispatches

district-wide SOS alerts.

🚜 Rule 2: Farmer Clustering Trigger

When ≥ 3 farmers in the same locality/taluka report matching clinical symptoms via multimodal scans or consult requests, an

active outbreak cluster is automatically formed with actionable biosecurity protocols.

🔄 Dynamic Outbreak Resolution Lifecycle

Recovery Reporting: Farmers or paravets submit recovery counts (`POST /api/outbreaks/report-recovery`). Active affected counts decrement dynamically.
Containment Clearance: When affected counts reach 0 or when veterinarians issue an official resolution certificate (`POST /api/outbreaks/resolve`), the outbreak transitions to `status = 'resolved'`, `severity = 'RESOLVED'`, the active red SOS flag is cleared, and an all-clear resolution notice is broadcast across the district.

4. End-to-End Architecture & Data Flow Diagram

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----------------------+ | PASHURAKSHAK FRONTEND SPA (React 18 / Tailwind) | | [Farmers: Herd Scanner &
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Outbreak SOS] [Vets: Case Queue & Duty Management] [Directory & Biosecurity Defense Radar] |
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+-------------------------------------------------------------+ | REST API (X-Auth-Token, X-User-ID,
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Multipart / JSON) -> http://localhost:8080 v
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----------------------+ | GO HIGH-PERFORMANCE BACKEND (Go 1.22+) | | - Authentication Middleware & Bcrypt
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Security - Outbreak Cluster Sensing & Consensus Engine | | - FFmpeg Temporal Keyframe Extractor (4 Frames) -
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Asynchronous Backup Snapshot Hook (VACUUM INTO) | +-----------------------------------
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+-------------------------------------------------+-------------------------------------+ | | Multipart
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Media Ingestion | AI Inference Request | Surveillance & Case Escalation v v v
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+-----------------------+ | STORAGE & MEDIA ENGINE | | DUAL-STAGE MULTIMODAL AI PIPELINE | | CONSENSUS &
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RECOVERY | | - Content-Addressable Storage (./uploads/) | | Stage 1: Qwen2.5-VL-72B-Instruct (Vision) | | -
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Vet Consensus ≥3| | - FFmpeg 4-Frame Keyframe Generator (JPEG) | | Stage 2: viggovet/viggoVet-Reasoning-20B
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| | - Farmer Cluster ≥3| +-----------------------------------------------+
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+-------------------------------------------------+ | TRIPLE SQLITE 3 WAL STORAGE | | - user.db: User
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accounts, Bcrypt hashes & auth | | - animal_health.db: Profiles, Aadhaar & Labs (v5) | | - surveillance.db:
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Outbreaks, SOS & Circulars | +-------------------------------------------------+
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5. Triple SQLite Schema Architecture & Data Flow Isolation

To ensure fault isolation and prevent surveillance queries from competing with clinic authentication, storage is split into three independent SQLite databases running in Write-Ahead Logging (WAL) mode:

Database & Engine	Table Name	Schema Structure & Key Indexes
`user.db` Schema v4 (WAL)	`users`	`id (PK)`, `name`, `email (UNIQUE)`, `password_hash`, `role`, `district`, `clinic_name`, `is_available`, `visiting_location`, `leave_notes`, `created_at`
	`sessions`	`token (PK)`, `user_id (INDEX)`, `role`, `expires_at`, `created_at`
	`user_profiles`	`user_id (PK)`, `farm_name`, `herd_size`, `village`, `taluka`, `district`, `clinic_hours`, `updated_at`
	`audit_logs`	`id (PK)`, `user_id`, `email`, `event`, `ip_address`, `details`, `created_at`
`animal_health.db` Schema v5 (WAL)	`animals`	`id (PK)`, `user_id`, `name`, `species`, `breed`, `age`, `sex`, `tag_number` (12-digit Pashu Aadhaar), `herd_size`, `village`, `taluka`, `district`, `photo_url`, `notes`, `weight`, `created_at`
	`health_screenings`	`id (PK)`, `animal_id`, `user_id`, `media_url`, `symptoms_json`, `visual_analysis_json`, `assessment_json`, `urgency`, `district`, `taluka`, `farm_name`, `vet_review_requested`, `created_at`
	`clinic_test_results`	`id (PK)`, `animal_id`, `user_id` (owner), `vet_id`, `vet_name`, `clinic_name`, `test_type`, `sample_date`, `test_parameters_json`, `interpretation`, `status` ('Normal'|'Abnormal'|'Critical'), `recommendation`, `created_at`
	`reminders`	`id (PK)`, `user_id`, `animal_id`, `type`, `title`, `description`, `due_at`, `completed`, `created_at`
`surveillance.db` Schema v2 (WAL)	`outbreaks`	`id (PK)`, `disease_name`, `species`, `district`, `taluka`, `village`, `farm_name`, `affected_count`, `severity`, `prevention_guide`, `status` ('active'|'contained'|'resolved'), `verified_by_vets`, `reported_by_farmers`, `resolution_notes`, `resolved_at`, `created_at`
	`notifications`	`id (PK)`, `user_id`, `role_target` ('all'|'farmer'|'vet'|'ngo'), `district`, `title`, `message`, `outbreak_id`, `severity`, `is_sos`, `read`, `action_url`, `created_at`
	`vet_consultations`	`id (PK)`, `screening_id`, `user_id`, `user_name`, `user_role`, `user_phone`, `location`, `vet_id`, `vet_name`, `animal_name`, `species`, `breed`, `media_url`, `symptoms_json`, `owner_notes`, `ai_diagnosis`, `ai_urgency`, `vet_diagnosis`, `vet_suggestion`, `vet_prescriptions`, `status` ('pending'|'reviewed'), `created_at`

Database & Engine	Table Name	Schema Structure & Key Indexes
	`inventories`	`id (PK)`, `user_id`, `owner_name`, `org_type`, `district`, `item_name`, `category`, `quantity`, `unit`, `min_threshold`, `expiry_date`, `status`, `updated_at`
	`gov_advisories`	`id (PK)`, `issuer`, `title`, `content`, `district`, `category`, `urgency`, `date_issued`, `created_at`

6. Complete RESTful API Specifications

Method & Path	Headers & Params	Description & Payload Architecture
`POST /api/auth/register`	`Content-Type: json`	Registers account with role (`farmer`, `pet_owner`, `vet`, `ngo`) and geo fields (farm name, village, taluka, clinic hours).
`POST /api/auth/login`	`Content-Type: json`	Authenticates credentials with Bcrypt verification; issues 30-day session token.
`POST /api/auth/profile`	`X-Auth-Token`	Updates profile, farm parameters, or veterinarian availability duty status & leave notices.
`POST /api/auth/change-password`	`X-Auth-Token`	Changes user password with old password verification and audit logging.
`GET /api/vets`	`?district=Pune`	Returns directory of licensed veterinarians with real-time duty status, visiting location, and clinic hours.
`POST /api/health-check/upload`	`multipart/form-data`	Validates and writes image/video to content-addressable storage.
`POST /api/health-check/analyze`	`X-Auth-Token`, `X-User-ID`	Executes 4-frame temporal extraction, visual AI inspection, and veterinary medical differential assessment.
`GET /api/outbreaks`	`?district=...&species=...`	Lists active and resolved disease clusters across Maharashtra districts.
`GET /api/outbreaks/:id`	Public / Auth	Fetches complete outbreak record with consensus stats, affected counts, and biosecurity guidelines.
`POST /api/outbreaks/report-recovery`	`Content-Type: json`	Reports herd recovery. Automatically decrements affected counts and clears SOS when resolved.
`POST /api/outbreaks/resolve`	`X-Auth-Token (Vet/Gov)`	Issues official veterinary clearance certificate and containment notice.
`GET /api/notifications`	`X-Auth-Token`	Fetches targeted and broadcast SOS alerts and biosecurity

Method & Path	Headers & Params	Description & Payload Architecture
		notices.
`POST /api/vet-consultations`	`X-Auth-Token`	Farmers/Pet Owners request second opinion on doubtful AI scans.
`POST /api/vet-consultations/review`	`X-Auth-Token (Vet)`	Doctor submits clinical diagnosis, advice, and ℞ prescriptions. Triggers consensus check.
`GET /api/animals`	`?q=...&tag_number=...`	Fetches user's animals (farmers/owners) or full regional directory with Pashu Aadhaar search (veterinarians).
`GET /api/animals/tag/:tag`	Public / Auth	Instant lookup of any livestock or companion animal by 12-digit Pashu Aadhaar ear tag.
`PUT /api/animals/:id`	`X-Auth-Token`	Updates animal profile details (species, breed, age, weight, Pashu Aadhaar tag, clinical notes).
`GET /api/clinic-test-results`	`?animal_id=...`	Confidential retrieval of doctor-issued lab test reports (accessible only by owner and attending vets).
`POST /api/clinic-test-results`	`X-Auth-Token (Vet)`	Veterinarian publishes official lab test report (CBC, Mastitis CMT, PCR) and notifies animal owner.
`GET /api/inventory`	`?district=...`	Returns dynamic stock of vaccines, antibiotics, antiseptics, and PPE supplies.
`PUT / DELETE /api/inventory`	`X-Auth-Token`	Updates or deletes inventory stock with strict ownership enforcement (only the creator can modify).
`GET /api/gov-advisories`	`?district=...`	Returns official Maharashtra animal husbandry circulars and ring vaccination directives.

7. Clinical Verification Benchmarks across Livestock & Companion Animals

Clinical Scenario	Visual Analysis (Qwen-VL)	Veterinary Differentials (ViggoVet)	Triage & Outbreak Status
Bovine Farm Cluster Baramati Dairy Farm, Cattle fever & salivation	Excessive ropy salivation, interdigital foot vesicles, oral erosions on tongue, lameness.	1. **[HIGH] Foot-and-Mouth Disease (FMD / लाळ्या खुरकूत)** 2. [MODERATE] Vesicular Stomatitis 3. [POSSIBLE] Bovine Viral Diarrhea (BVD)	CRITICAL SOS Verified Outbreak Active

Clinical Scenario	Visual Analysis (Qwen-VL)	Veterinary Differentials (ViggoVet)	Triage & Outbreak Status
Caprine Epidemic Sangamner Goat Herd, fever & erosions	Mucopurulent ocular-nasal discharge, necrotic stomatitis, dyspnea, diarrhea staining.	1. **[HIGH] Peste des Petits Ruminants (PPR)** 2. [MODERATE] Contagious Caprine Pleuropneumonia 3. [POSSIBLE] Goat Pox	URGENT Ring Vaccine Deployed
Canine Motion Clip 10s video of puppy twitching	Rhythmic involuntary myoclonus (chewing-gum seizures), nasal hyperkeratosis.	1. **[HIGH] Canine Distemper Encephalomyelitis** 2. [MODERATE] Idiopathic Epilepsy 3. [POSSIBLE] Viral Meningoencephalitis	EMERGENCY Doctor Consult Sent
Healed Dairy Herd Malegaon Budruk, Day 14 Follow-up	Complete re-epithelialization of oral lesions, normal gait, voluntary feeding resumed.	1. **[RESOLVED] Post-FMD Clinical Recovery** 2. Complete tissue healing confirmed.	RESOLVED SOS Cleared • Movement Open